IMPERIAL COLLEGE LONDON

EE4-57
EE9-CS2-2

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE/EIE PART IV: MEng and ACGI

DISCRETE-EVENT SYSTEMS

Wednesday, 6 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : D. Angeli
Second Marker(s) : E.C. Kerrigan

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1. A storage facility of total capacity 6 is used to stockpile two types of boxes, of size 2 and 3, respectively. Each time a box of the first kind arrives and fits in the store an event a_1 is generated, while an event d_1 corresponds to a box of the first kind being dispatched from the store. Similarly event a_2 represents arrivals of boxes of the second kind, while d_2 corresponds to their departure from the store.

   a) Assuming the storage is initially empty, build a finite deterministic automaton G, which is able to keep track of the number of boxes of type 1 and 2 inside the store, as well as of its capacity limits. [4]

   b) Occasionally boxes of the second kind (size 3) are misclassified, meaning that, provided there is room in the storage, these are allowed in but an event $\tilde{a}_1$ is generated instead. Build a deterministic automaton $\tilde{G}$ where the (true) number of boxes of type 1 and 2 is kept track of and where events $\tilde{a}_1$ are included. [3]

   c) Consider next a labelling automaton G_L with 2 states $\{N,Y\}$, where N is the initial state, and the single event $\tilde{a}_1$ (same event as in item b)), such that $\tilde{a}_1$ in N triggers the transition to Y, while $\tilde{a}_1$ in Y leads to the same state Y. Compute the parallel composition $G_P = \tilde{G}||G_L$. [4]

   d) Assume next that events $\tilde{a}_1$ and a_1 are partially observable, meaning that they are recorded merely as a_1 events. Build a non-deterministic automaton G_N by replacing $\tilde{a}_1$ events with a_1 events in G_P. Synthesize a diagnoser capable of detecting occurrence of $\tilde{a}_1$ events, by building the Observer automaton of G_N. (Suggestion: for the sake of simplicity collapse all states where only Y labels appear as a single Y state). [6]

   e) The diagnoser classifies strings as N (no $\tilde{a}_1$ events occured), (Y, $\tilde{a}_1$ events occurred) or (YN, meaning both possibilities are open). How is your diagnoser classifying each of the following three strings : $a_1a_1a_1d_1$, $a_2a_1d_2d_2$, a_2a_1 ? Could you provide an intuitive explanation of why this is the case for each string? [3]

2. Two machines (M1 and M2) work in parallel and produce 2 types of pieces, P1 and P2, respectively. Each machine operates between 2 states, Busy and Idle, and commutes between them according to events p_i (produced piece of type i) and d_i (delivered piece of type i, after which the machine can restart operations and is Busy again). A Robot, enabled to carry a single piece at a time, picks them up from the machine i (event d_i) and unloads them (event u_i) at the assembling machine M_A.

a) Build finite deterministic automata G_1 and G_2 to model the individual machines. Let the Busy state be their initial and terminal state. [3]

b) Build a finite deterministic automaton G_R to model the Robot, and the type of piece it is carrying. Assume the robot is initially not carrying any pieces (Empty state) and that the empty state is the only terminal state. [3]

c) Compute the parallel composition G of the models G_1, G_2 and G_R. [4]

d) Model the assembling machine as a specification automaton. In particular, the following constraints need to be met: the machine accepts pieces of both types but needs exactly one piece of type 1 and one of type 2 to be able to start the assembling operations and successfully complete them. There are no buffers at the machine, so pieces have to come in the right amounts. The order in which pieces come is not essential (viz. the machine is able to collect pieces from the robot in any order). Let G_{spec} be the corresponding automaton, with event set $\{u_1, u_2\}$. Assume the machine is initially empty and its initial state is also its terminal state. [3]

e) Assume that all events are controllable. Describe the procedure to design a non-blocking supervisor that implements the specification and whose closed-loop language is maximal. [3]

f) Sketch the automaton G_S implementing the supervisor $S(\cdot)$ described above, in particular $S(s) = \Gamma_{G_S}(f_{G_S}(x_0, s))$. [4]

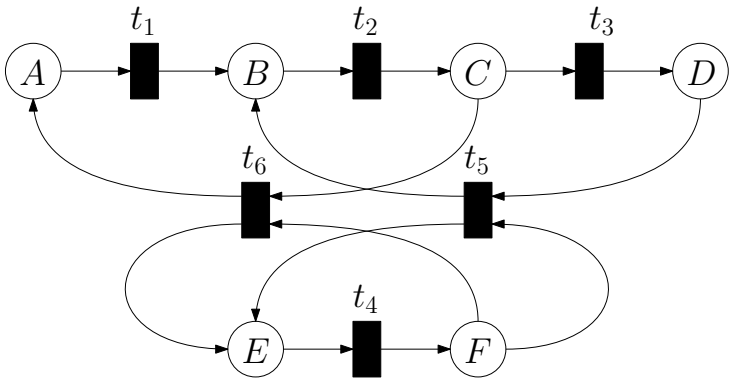


Figure 3.1 Petri Net model of chemical network

3. A network of chemical reactions (CRN), can be modeled as a Petri Net. Below we represent the following set of reactions:

$$\begin{array}{rcl} A \to B & \to & C \to D \\ E & \to & F \\ D+F & \to & B+E \\ C+F & \to & A+E, \end{array}$$

involving chemical species A, B, C, D, E and F with the Petri Net N shown in Fig. 3.1.

a) Find the incidence matrix associated to the Petri Net in Fig. 3.1. [4]

b) Find the P-semiflows of minimal support of N and provide an interpretation of their physical meaning. [4]

c) Comment on the structural boundedness and/or conservativity of the network N. [2]

d) Find the T-semiflows of N and comment on their physical meaning. [4]

e) Consider the initial marking $M_0 = [1,0,0,0,1,0]'$. Find the reachable set $\mathscr{R}(M_0)$ and sketch the corresponding transition diagram. [4]

f) Is the marked Petri Net (N, M_0) reversible? (justify your answer) [2]

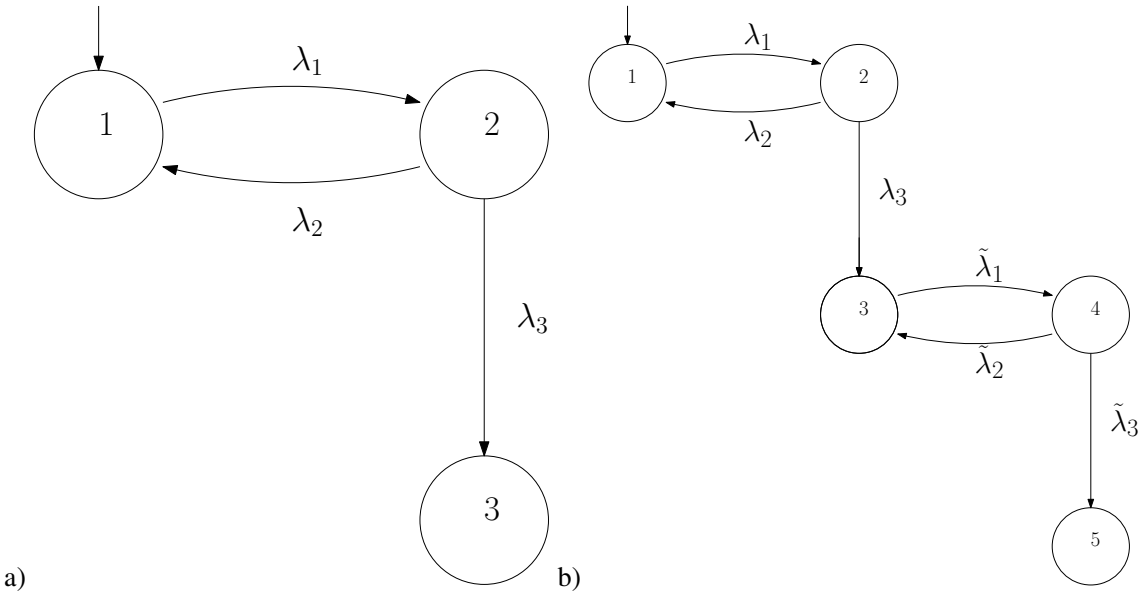


Figure 4.1 Transition diagrams of the Markov Chains

4. Consider the continuous time Markov chain associated with the transition diagram shown in Fig. 4.1 a). Assume that the positive rates of transition probabilities are λ_1, λ_2 and λ_3 as from the associated diagram.

   a) Write the Kolmogorov equations for the evolution of probabilities $\pi_i(t) := Pr\{x(t) = i\}$, $i = 1,2,3$ for the considered Markov chain. [4]

   b) Is the chain ergodic? (justify your answer) [2]

   c) Find the transfer function $G_1(s)$ from state 1 to output $y(t) = \lambda_3 \pi_2(t)$. [3]

   d) Use $G_1(s)$ to compute the expected value of absorption time for state 3 when the initial state is 1 (with probability 1). [3]

   e) Consider next the Markov Chain shown in Fig. 4.1 b). For the associated Kolmogorov equations compute the transfer function $G(s)$ between state 1 and output $y = \tilde{\lambda}_3 \pi_4(t)$. (*Hint: to simplify calculations interpret it as a cascade of two linear systems*) [4]

   f) Exploiting the factorised expression of the transfer function $G(s)$, show that the expected value of the absorption time for state 5 of the cascaded Markov chain from initial state 1 is the sum of the absorption times of two simpler Markov chains as discussed in d). [4]